The "Negative-Viscosity" Effect in a Magnetic Fluid: First-Principles Derivations of the Complete Theory

Bacri, Perzynski, Shliomis, Burde, PRL 75, 2128 (1995), built up from its sources:
Shliomis, Sov. Phys. JETP 34, 1291 (1972) [Ref. 6];
Martsenyuk, Raikher, Shliomis, Sov. Phys. JETP 38, 413 (1974) [Ref. 8];
Shliomis & Morozov, Phys. Fluids 6, 2855 (1994) [Ref. 5];
Shliomis & Stepanov, JMMM 122, 176 (1993); McTague, JCP 51, 133 (1969) [Ref. 7]

Derivation notes. Every boxed equation is re-derived here and checked (limits, symmetries, or direct numerics); see the Verification Ledger, Sec. 15.

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1 Conventions, symbols, and the plan

Gaussian (CGS) units throughout, as in all the source papers.

Symbol	Units	Meaning
η	P	zero-field viscosity of the suspension (experiment: 77 cP)
$\Delta\eta$	P	rotational (vortex) viscosity contributed by the field
$\varphi = nV$	—	solid volume fraction (0.20); n = particle number density
d, V, a		particle diameter (10 nm), volume $V = \pi d^3/6$, radius $a = d/2$
ρ_s	g/cm^3	density of the grain material
M_s, $m = M_s V$	G, erg/G	grain spontaneous magnetization (400 G); particle moment
K, $\sigma = KV/kT$	erg/cm^3	uniaxial anisotropy constant (2×10^6); barrier parameter
$\mathbf{e}$	—	unit vector along the particle magnetic moment, $\mathbf{m} = m\mathbf{e}$
$\mathbf{H} = H(t)\mathbf{h}$	Oe	applied field; in the PRL $H(t) = H_0 \cos\omega t$, $\mathbf{h}$ along the tube
$\xi = mH/kT$	—	Langevin parameter; amplitude value $\xi_0 = mH_0/kT$; rms $\bar{\xi} = \xi_0/\sqrt{2}$
$L(\xi) = \coth\xi - 1/\xi$	—	Langevin function; $L'(\xi) = \xi^{-2} - \sinh^{-2}\xi$
$\mathbf{M}$	G	magnetization; $\mathbf{M} = nm\langle\mathbf{e}\rangle$
$\boldsymbol{\Omega} = \frac{1}{2}\nabla \times \mathbf{v}$	s^{-1}	flow vorticity (half the curl, as in all source papers)
$\boldsymbol{\Theta} \equiv \boldsymbol{\omega}_p$	s^{-1}	mean angular velocity of the particles
$\zeta_r = 6\eta V$	erg s	rotational friction coefficient of one particle
$D_r = kT/\zeta_r$	s^{-1}	rotational diffusion coefficient
$\tau_B = 1/2D_r = 3\eta V/kT$	s	Brownian orientational relaxation time (measured 1.6 ms)
τ_N	s	Néel (internal) relaxation time
τ_s, I	s, g/cm	spin-inertia decay time; particles' moment of inertia per volume
$g(\xi_0, \omega\tau_B)$	—	normalized rotational viscosity: $\Delta\eta = \frac{3}{2}\eta\varphi\, g$
$\hat{\mathbf{R}} = \mathbf{e} \times \partial/\partial\mathbf{e}$	—	infinitesimal-rotation operator on the unit sphere

Plan. Part A (Secs. 2–5): statistical mechanics of one rigid dipole — Fokker–Planck equation, exact moment equation, and the three magnetization closures (Debye; Shliomis 1972; effective field of Martsenyuk–Raikher–Shliomis, "MRSh"). Part B (Secs. 6–7): ferrohydrodynamics with internal rotation and the general rotational-viscosity theorem. Part C (Secs. 8–10): the three field regimes solved — static (McTague/Shliomis), weak AC (Shliomis–Morozov Sec. II), arbitrary-amplitude AC with the Debye closure (SM Sec. III, including a corrected Eq. (40) — see Sec. 16). Part D (Secs. 11–13): the PRL's own theory (MRSh + perturbation in $\Omega\tau_B$), the numerics, and the experiment. Sec. 15: verification ledger. Sec. 16: errata found while verifying.

Part I
Statistical mechanics of a rigid dipole

2 Single-particle physics

2.1 Rotational friction and the two microscopic times

Low-Reynolds rotation of a sphere of radius a at angular velocity $\boldsymbol{\omega}$ relative to the surrounding liquid produces the Stokes torque

$$\mathbf{T}_{\rm fr} = -8\pi\eta a^3\boldsymbol{\omega} = -6\eta V\boldsymbol{\omega} \quad \Rightarrow \quad \zeta_r = 6\eta V. \tag{1}$$

Inertial time. A free particle's spin obeys $I_1\dot{\boldsymbol{\omega}} = -\zeta_r\boldsymbol{\omega}$ with $I_1 = \rho_s V d^2/10$, so perturbations of $\boldsymbol{\omega}$ decay in

$$\tau_s = \frac{I_1}{\zeta_r} = \frac{\rho_s V d^2/10}{6\eta V} = \frac{\rho_s d^2}{60\eta} \approx \frac{6 \times (10^{-6})^2}{60 \times 10^{-2}}\ \text{s} = 10^{-11}\ \text{s}, \tag{2}$$

using $d = 10^{-6}$ cm, $\rho_s = 6$ g/cm^3, $\eta = 10^{-2}$ g/cm s [SM Eq. (1)]. *Diffusional time.* The Einstein relation gives $D_r = kT/\zeta_r$; the $l = 1$ orientational harmonic (i.e. $\langle \mathbf{e} \rangle$) relaxes at the rate $l(l+1)D_r = 2D_r$ (proved in Sec. 4), hence

$$\boxed{\tau_B = \frac{1}{2D_r} = \frac{3\eta V}{kT}}. \tag{3}$$

Since $\tau_s/\tau_B = \rho_s d^2 kT/(180\eta^2 V) \sim 10^{-5}$, spin inertia is irrelevant on every relevant time scale; this hierarchy is used repeatedly.

2.2 Néel vs. Brownian relaxation; the rigid-dipole regime

The moment $\mathbf{m}$ can relax (i) with the particle body (Brownian, time τ_B) or (ii) inside the body over the anisotropy barrier KV (Néel, time $\tau_N \simeq \tau_0 e^{\sigma}$, $\tau_0 \sim 10^{-9}$ s, $\sigma = KV/kT$). The two channels act in parallel on the orientation of $\mathbf{m}$, so the rates add:

$$\tau^{-1} = \tau_N^{-1} + \tau_B^{-1}, \tag{4}$$

which is Shliomis–Stepanov Eq. (7) specialized to the isotropic combination [their paper refines this into distinct longitudinal/transverse times $\tau_\parallel, \tau_\perp$ that also depend on σ; we only need the parallel-channels logic here]. For the Co-ferrite grains of the PRL,

$$\sigma = \frac{KV}{kT} = \frac{2 \times 10^6 \times 5.24 \times 10^{-19}}{4.1 \times 10^{-14}} \approx 26 \;\Rightarrow\; \tau_N \sim \tau_0 e^{26} \sim 10^2 \text{ s} \gg \tau_B, \tag{5}$$

so the Néel channel is frozen: the dipole is rigid, and *any* change of $\mathbf{M}$ requires mechanical particle rotation. The moment stays locked up to fields of order the anisotropy field $H_a \sim 2K/M_s = 10^4$ Oe, exactly the PRL's statement. (Shliomis–Stepanov's further point — that in polydisperse fluids the interplay of $\tau_N(V) \propto e^{\sigma}$ [steep] and $\tau_B(V) \propto V$ [mild] produces the susceptibility maxima and $1/\omega$ noise — is background for why one must check which mechanism operates before applying any of the following.)

2.3 Equilibrium averages needed later

With orientational energy $U = -\mathbf{m}\cdot\mathbf{H} = -kT\,\xi(\mathbf{e}\cdot\mathbf{h})$ the Gibbs distribution is $W_0(\mathbf{e}) \propto e^{\xi(\mathbf{e}\cdot\mathbf{h})}$. Partition function over the unit sphere:

$$Z(\xi) = \int e^{\xi\cos\theta}\,\mathrm{d}\Omega_{\mathbf{e}} = 2\pi \int_{-1}^{1} e^{\xi u}\mathrm{d}u = \frac{4\pi\sinh\xi}{\xi}. \tag{6}$$

First moment:

$$\langle\cos\theta\rangle_0 = \frac{\mathrm{d}\ln Z}{\mathrm{d}\xi} = \coth\xi - \frac{1}{\xi} = L(\xi) \;\Rightarrow\; \boxed{\mathbf{M}_0 = nm\,L(\xi)\,\mathbf{h} = \varphi M_s L(\xi)\mathbf{h}}. \tag{7}$$

Second moment: $\langle\cos^2\theta\rangle_0 = Z''/Z$. Using $Z' = ZL$, $Z'' = Z(L^2 + L')$, and the identity

$$L' = 1 - \coth^2\xi + \xi^{-2} \;\Rightarrow\; L^2 + L' = 1 - \frac{2}{\xi}\Big(\coth\xi - \frac{1}{\xi}\Big) = 1 - \frac{2L}{\xi}, \tag{8}$$

(each step is pure algebra; verified also numerically to 10^{-6})

$$\langle\cos^2\theta\rangle_0 = 1 - \frac{2L(\xi)}{\xi}, \qquad \langle\sin^2\theta\rangle_0/2 = \frac{L(\xi)}{\xi}. \tag{9}$$

Hence the full second-moment tensor under a Gibbs distribution aligned with unit vector $\hat{\boldsymbol{\nu}}$ at parameter ξ:

$$\boxed{\langle\mathbf{ee}\rangle_0 = \frac{L}{\xi}\,\mathbb{1} + \Big(1 - \frac{3L}{\xi}\Big)\hat{\boldsymbol{\nu}}\hat{\boldsymbol{\nu}}} \qquad (\text{trace} = \tfrac{3L}{\xi} + 1 - \tfrac{3L}{\xi} = 1\ \checkmark). \tag{10}$$

Limits: $\xi \to 0$: $\langle\mathbf{ee}\rangle \to \mathbb{1}/3$ (isotropy); $\xi \to \infty$: $\to \hat{\boldsymbol{\nu}}\hat{\boldsymbol{\nu}}$ (full alignment). Both correct.

3 The rotational Fokker–Planck equation

3.1 Deterministic drift of one particle in flow and field

Neglecting inertia (justified by τ_s), the instantaneous torque balance on one particle reads

$$0 = \underbrace{\mathbf{m}\times\mathbf{H}}_{\text{magnetic}} + \underbrace{(-\zeta_r)(\boldsymbol{\omega}-\boldsymbol{\Omega})}_{\text{Stokes, Eq. (1)}} + \mathbf{T}_{\text{th}}(t), \tag{11}$$

so the deterministic part of the particle's angular velocity is

$$\boldsymbol{\omega}_{\text{det}} = \boldsymbol{\Omega} + \frac{1}{\zeta_r}\,\mathbf{m}\times\mathbf{H}, \qquad \dot{\mathbf{e}} = \boldsymbol{\omega}_{\text{det}}\times\mathbf{e} + (\text{noise}). \tag{12}$$

Note that the magnetic torque can be written with the rotation operator $\hat{\mathbf{R}} = \mathbf{e}\times\partial/\partial\mathbf{e}$:

$$\mathbf{m}\times\mathbf{H} = -\hat{\mathbf{R}}U, \qquad U = -\mathbf{m}\cdot\mathbf{H}. \tag{13}$$

Check of (13)*:* $\hat{\mathbf{R}}U = \mathbf{e}\times\partial_{\mathbf{e}}(-mH\,\mathbf{e}\cdot\mathbf{h}) = -mH\,\mathbf{e}\times\mathbf{h} = -\mathbf{m}\times\mathbf{H}$. ✓

3.2 The kinetic equation

Let $W(\mathbf{e},t)$ be the orientation distribution ($\int W\,\mathrm{d}\Omega_{\mathbf{e}} = 1$). Probability conservation on the sphere, with drift (12) and rotational diffusion D_r (fluctuation–dissipation pair of the Stokes friction), gives the rotational Smoluchowski (Fokker–Planck) equation, first written for this problem by Brown and used by MRSh:

$$\boxed{\frac{\partial W}{\partial t} = \hat{\mathbf{R}}\cdot\Big[D_r\Big(\hat{\mathbf{R}}W + \frac{W}{kT}\,\hat{\mathbf{R}}U\Big) - \boldsymbol{\Omega}\,W\Big]} \tag{14}$$

Reading: the flux on the sphere is $\mathbf{J} = -D_r(\hat{\mathbf{R}}W + W\hat{\mathbf{R}}U/kT) + \boldsymbol{\Omega}W$; the first piece is diffusion, the second the potential drift (ζ_r^{-1} torque)W [consistent because $D_r/kT = 1/\zeta_r$], the third rigid convection by the vorticity.

Consistency checks. (i) Equilibrium: for $\boldsymbol{\Omega} = 0$, $W_0 \propto e^{-U/kT}$ annihilates the bracket exactly: $\hat{\mathbf{R}}W_0 = -W_0\hat{\mathbf{R}}U/kT$. ✓ (ii) $\boldsymbol{\Omega}$-convection conserves any isotropic distribution: $\hat{\mathbf{R}}\cdot(\boldsymbol{\Omega}W) = \boldsymbol{\Omega}\cdot\hat{\mathbf{R}}W = 0$ if W isotropic. ✓ (iii) Normalization: $\int\hat{\mathbf{R}}\cdot(\dots) = 0$ on a closed sphere. ✓

3.3 Operator identities (used below; each verified by index algebra)

With $(\hat{\mathbf{R}})_j = \varepsilon_{jkl}e_k\partial/\partial e_l$:

$$\hat{\mathbf{R}}_j e_i = \varepsilon_{jki}e_k, \tag{15}$$

$$\hat{\mathbf{R}}^2 e_i = \hat{\mathbf{R}}_j(\varepsilon_{jki}e_k) = \varepsilon_{jki}\,\varepsilon_{jlk}\,e_l = (\delta_{kl}\delta_{ik} - \delta_{kk}\delta_{il})\,e_l = e_i - 3e_i = -2e_i, \tag{16}$$

$$(\hat{\mathbf{R}}e_i)\cdot(\mathbf{e}\times\mathbf{h}) = \varepsilon_{jki}e_k\,\varepsilon_{jlm}e_l h_m = (\delta_{kl}\delta_{im} - \delta_{km}\delta_{il})e_k e_l h_m = h_i - e_i(\mathbf{e}\cdot\mathbf{h}), \tag{17}$$

where (16) uses $\varepsilon_{jki}\varepsilon_{jlk} = \delta_{kl}\delta_{ik}\cdot(\dots)$; the clean way: e_i is a linear combination of $l=1$ spherical harmonics and $\hat{\mathbf{R}}^2 = -\hat{L}^2$ has eigenvalue $-l(l+1) = -2$ on them. Identity (17) is the direct contraction shown. All three were verified symbolically.

4 The exact first-moment (magnetization) equation

Multiply (14) by e_i, integrate over the sphere, and integrate by parts ($\int f\,\hat{\mathbf{R}}\cdot\mathbf{A} = -\int(\hat{\mathbf{R}}f)\cdot\mathbf{A}$ on a closed sphere):

$$\frac{\mathrm{d}\langle e_i\rangle}{\mathrm{d}t} = -\int(\hat{\mathbf{R}}e_i)\cdot\Big[D_r\hat{\mathbf{R}}W + \frac{D_r W}{kT}\hat{\mathbf{R}}U - \boldsymbol{\Omega}W\Big]\mathrm{d}\Omega. \tag{18}$$

Term by term:

- *Diffusion:* $-D_r \int (\hat{\mathbf{R}} e_i)\cdot \hat{\mathbf{R}} W = +D_r \int (\hat{\mathbf{R}}^2 e_i) W = -2D_r \langle e_i\rangle$ by (16).
- *Potential drift:* $\hat{\mathbf{R}} U/kT = -\xi\, \mathbf{e}\times\mathbf{h}$ by (13); then by (17), $-\int (\hat{\mathbf{R}} e_i)\cdot\big(-\xi D_r(\mathbf{e}\times\mathbf{h})\big)W = \xi D_r \langle h_i - e_i(\mathbf{e}\cdot\mathbf{h})\rangle$.
- *Convection:* $+\int (\hat{\mathbf{R}} e_i)\cdot \boldsymbol{\Omega}\, W = \varepsilon_{jki}\Omega_j \langle e_k\rangle = (\boldsymbol{\Omega}\times\langle\mathbf{e}\rangle)_i$.

Using $2D_r = 1/\tau_B$:

$$\boxed{\frac{\mathrm{d}\langle\mathbf{e}\rangle}{\mathrm{d}t} = \boldsymbol{\Omega}\times\langle\mathbf{e}\rangle - \frac{1}{\tau_B}\langle\mathbf{e}\rangle + \frac{\xi}{2\tau_B}\big(\mathbf{h} - \langle\mathbf{ee}\rangle\cdot\mathbf{h}\big)} \tag{19}$$

This equation is *exact* — no approximation yet — but not closed: it involves $\langle\mathbf{ee}\rangle$, whose equation involves the third moment, etc. (the classical moment hierarchy).

Verification against equilibrium. Insert $\langle\mathbf{e}\rangle = L\mathbf{h}$, $\langle\mathbf{ee}\rangle$ from (10) with $\hat{\boldsymbol{\nu}} = \mathbf{h}$, $\boldsymbol{\Omega} = 0$: $\langle\mathbf{ee}\rangle\cdot\mathbf{h} = \big[\frac{L}{\xi} + 1 - \frac{3L}{\xi}\big]\mathbf{h} = \big(1 - \frac{2L}{\xi}\big)\mathbf{h}$, so the RHS is $-\frac{L}{\tau_B}\mathbf{h} + \frac{\xi}{2\tau_B}\cdot\frac{2L}{\xi}\mathbf{h} = 0$. ✓ Stationarity of the Gibbs state is reproduced identically — a nontrivial check of every coefficient in (19).

5 Three closures of the moment hierarchy

5.1 Closure 1: near-isotropy ⇒ the Debye equation (exact to $O(\xi)$)

If the distribution is weakly anisotropic (weak field, i.e. $\xi \ll 1$, arbitrary time dependence), $\langle\mathbf{ee}\rangle = \mathbb{1}/3 + O(\xi)$, and the last term of (19) becomes $\frac{\xi}{2\tau_B}\cdot\frac{2}{3}\mathbf{h} = \frac{\xi}{3\tau_B}\mathbf{h}$. With $\mathbf{M} = nm\langle\mathbf{e}\rangle$ and $\chi = nm^2/3kT$:

$$\boxed{\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t} = \boldsymbol{\Omega}\times\mathbf{M} - \frac{1}{\tau_B}\big(\mathbf{M} - \chi\mathbf{H}(t)\big)} \qquad (\xi \ll 1,\ \text{any } \omega). \tag{20}$$

So the Debye relaxation law is not a postulate; it is the *exact* weak-field limit of the Fokker–Planck dynamics. This underwrites all of Sec. 9.

5.2 Closure 2: Shliomis (1972) phenomenological extension

Shliomis extended (20) to finite ξ by replacing $\chi\mathbf{H} \to \mathbf{M}_0(t) = \varphi M_s L(\xi(t))\mathbf{h}$ and convecting with the *particle* rotation $\boldsymbol{\Theta}$ (which, unlike $\boldsymbol{\Omega}$, includes the back-action of the magnetic torque):

$$\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t} = \boldsymbol{\Theta}\times\mathbf{M} - \frac{1}{\tau_B}\big(\mathbf{M} - \mathbf{M}_0(t)\big) \tag{SM 7}$$

Its exactness domain: (a) $\xi \ll 1$ (reduces to (20) since then $\boldsymbol{\Theta}$–$\boldsymbol{\Omega}$ difference enters at higher order); (b) *static* $\mathbf{H}$ of any strength — because then the stationary state of (SM 7) in a quiescent fluid is $\mathbf{M} = \mathbf{M}_0$, which coincides with the exact Gibbs result (7). For finite ξ *and* finite $\omega\tau_B$ it is uncontrolled — this is the precise content of the "discrepancy" paragraph of the PRL and the reason Sec. 5.3 exists.

5.3 Closure 3: the effective-field (MRSh) closure ⇒ PRL Eqs. (5),(6)

Ansatz (Martsenyuk–Raikher–Shliomis 1974): at every instant the true nonequilibrium W is replaced by a Gibbs form in a fictitious dimensionless field $\boldsymbol{\xi}_e(t)$:

$$W_e(\mathbf{e},t) \propto \exp\big(\boldsymbol{\xi}_e(t)\cdot\mathbf{e}\big) \quad\Rightarrow\quad \langle\mathbf{e}\rangle = L(\xi_e)\,\hat{\boldsymbol{\xi}}_e, \qquad \boxed{\mathbf{M} = \varphi M_s L(\xi_e)\,\frac{\boldsymbol{\xi}_e}{\xi_e}} \tag{PRL 5}$$

and, crucially, the second moment is evaluated with the *same* W_e via (10) with $\hat{\boldsymbol{\nu}} = \hat{\boldsymbol{\xi}}_e$, parameter ξ_e:

$$\langle\mathbf{ee}\rangle_e\cdot\mathbf{h} = \frac{L_e}{\xi_e}\,\mathbf{h} + \Big(1 - \frac{3L_e}{\xi_e}\Big)(\hat{\boldsymbol{\xi}}_e\cdot\mathbf{h})\,\hat{\boldsymbol{\xi}}_e. \tag{21}$$

Substitute into the exact moment equation (19):

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(L_e\hat{\boldsymbol{\xi}}_e\big) = \boldsymbol{\Omega}\times\big(L_e\hat{\boldsymbol{\xi}}_e\big) - \frac{L_e}{\tau_B}\hat{\boldsymbol{\xi}}_e + \frac{\xi}{2\tau_B}\Big[\Big(1-\frac{L_e}{\xi_e}\Big)\mathbf{h} - \Big(1-\frac{3L_e}{\xi_e}\Big)(\hat{\boldsymbol{\xi}}_e\cdot\mathbf{h})\hat{\boldsymbol{\xi}}_e\Big]. \tag{22}$$

Claim: (22) is identical to the PRL's Eq. (6),

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\Big(L_e\frac{\boldsymbol{\xi}_e}{\xi_e}\Big) = \boldsymbol{\Omega}\times\Big(L_e\frac{\boldsymbol{\xi}_e}{\xi_e}\Big) - \frac{1}{\tau_B}\frac{L_e}{\xi_e}(\boldsymbol{\xi}_e-\boldsymbol{\xi}) - \frac{1}{2\tau_B\xi_e^2}\Big(1-\frac{3L_e}{\xi_e}\Big)\,\boldsymbol{\xi}_e\times(\boldsymbol{\xi}_e\times\boldsymbol{\xi})} \tag{PRL 6}$$

with $\boldsymbol{\xi}=\xi\mathbf{h}$ the true field. *Proof by expansion of* (PRL 6)*:*

$$-\frac{L_e}{\tau_B\xi_e}(\boldsymbol{\xi}_e-\boldsymbol{\xi}) = -\frac{L_e}{\tau_B}\hat{\boldsymbol{\xi}}_e + \frac{L_e\,\xi}{\tau_B\,\xi_e}\mathbf{h}, \tag{23}$$

$$\boldsymbol{\xi}_e\times(\boldsymbol{\xi}_e\times\boldsymbol{\xi}) = \boldsymbol{\xi}_e(\boldsymbol{\xi}_e\cdot\boldsymbol{\xi}) - \boldsymbol{\xi}\,\xi_e^2 = \xi_e^2\xi\big[(\hat{\boldsymbol{\xi}}_e\cdot\mathbf{h})\hat{\boldsymbol{\xi}}_e - \mathbf{h}\big], \tag{24}$$

$$\Rightarrow\ \text{3rd term} = \frac{\xi}{2\tau_B}\Big(1-\frac{3L_e}{\xi_e}\Big)\big[\mathbf{h}-(\hat{\boldsymbol{\xi}}_e\cdot\mathbf{h})\hat{\boldsymbol{\xi}}_e\big]. \tag{25}$$

Collect the $\mathbf{h}$ coefficients: $\frac{\xi}{\tau_B}\Big[\frac{L_e}{\xi_e}+\frac{1}{2}-\frac{3L_e}{2\xi_e}\Big] = \frac{\xi}{2\tau_B}\Big(1-\frac{L_e}{\xi_e}\Big)$, and the $(\hat{\boldsymbol{\xi}}_e\cdot\mathbf{h})\hat{\boldsymbol{\xi}}_e$ coefficients: $-\frac{\xi}{2\tau_B}\big(1-\frac{3L_e}{\xi_e}\big)$ — exactly (22). ■

So PRL Eq. (6) is nothing mysterious: it is the exact first-moment equation (19) closed with the Gibbs-form second moment (10). Its exactness domains: equilibrium (trivially), weak fields (where it reduces to (20) — check: $\xi_e\ll 1$, $L_e\to\xi_e/3$, $1-3L_e/\xi_e\to O(\xi_e^2)$, third term drops, second term $\to-(\boldsymbol{\xi}_e-\boldsymbol{\xi})/3\tau_B$, i.e. Debye), and it is known from MRSh's comparison with direct numerical solution of (14) to be accurate to a few percent everywhere. Note there is *no* separate $\mathbf{M}\times(\mathbf{M}\times\mathbf{H})$ term here: the magnetic-torque drift is already inside the potential term of (14).

5.4 Linearized relaxation times of the MRSh equation (used to interpret everything)

Linearize (PRL 6) about a constant field $\boldsymbol{\xi}=\xi\mathbf{h}$. (i) *Longitudinal* ($\boldsymbol{\xi}_e=(\xi+\delta\xi)\mathbf{h}$): projecting on $\mathbf{h}$, $L'\dot{\delta\xi} = -\frac{L}{\tau_B\xi}\delta\xi$, so

$$\tau_\parallel = \tau_B\,\frac{\xi L'(\xi)}{L(\xi)} = \tau_B\,\frac{\mathrm{d}\ln L}{\mathrm{d}\ln\xi} \quad\to\quad \tau_B\ (\xi\ll1),\qquad \tau_B/\xi\ (\xi\gg1). \tag{26}$$

(ii) *Transverse* ($\boldsymbol{\xi}_e=\xi\mathbf{h}+\boldsymbol{\delta}$, $\boldsymbol{\delta}\perp\mathbf{h}$): from the algebra done in Sec. 8.2 below, the restoring rate is $\frac{1}{2\tau_B}(1-L/\xi)$, i.e.

$$\tau_\perp = \frac{2\tau_B}{1-L/\xi} = \frac{2\tau_B\,\xi}{\xi-L} \quad\to\quad \tau_B\ (\xi\ll1),\qquad \frac{2\tau_B}{\xi}\ (\xi\gg1). \tag{27}$$

Checks: both reduce to τ_B at $\xi\to0$ (isotropy: no distinction) ✓; both shorten in strong fields (stiff restoring torque) ✓. The Debye closure has one time τ_B for everything — *this* is what fails at finite ξ, $\omega\tau_B$.

Part II
Ferrohydrodynamics with internal rotation

6 The coupled field–flow equations (SM Eqs. 6–10 → PRL Eqs. 1,3,4)

6.1 Magnetic body force (Kelvin force)

A dipole $\mathbf{m}$ in a nonuniform field feels $\mathbf{F}=(\mathbf{m}\cdot\nabla)\mathbf{H}$ (gradient of $-U$ at fixed $\mathbf{m}$, using $\nabla\times\mathbf{H}=0$ inside a nonconducting fluid). Per unit volume: $(\mathbf{M}\cdot\nabla)\mathbf{H}$. Together with the magnetostatic

constraints for a nonconducting, incompressible medium,

$$\nabla\cdot\mathbf{v}=0, \qquad \nabla\times\mathbf{H}=0, \qquad \nabla\cdot(\mathbf{H}+4\pi\mathbf{M})=0, \tag{SM 9}$$

this fixes the "field" part of the momentum equation. (In the PRL geometry $\mathbf{H}$ is uniform along the tube, so $(\mathbf{M}\cdot\nabla)\mathbf{H}=0$; it is kept for completeness.)

6.2 Internal angular momentum and the antisymmetric stress

Let $\mathbf{S}=I\boldsymbol{\Theta}$ be the internal (spin) angular momentum density. Its balance:

$$I\,\frac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}t}=\mathbf{M}\times\mathbf{H}-\frac{I}{\tau_s}(\boldsymbol{\Theta}-\boldsymbol{\Omega}), \qquad \frac{I}{\tau_s}=6\eta\varphi, \tag{SM 8}$$

where the identification $I/\tau_s=6\eta\varphi$ is checked directly: $I=n\,\frac{1}{10}\rho_s V d^2=\varphi\rho_s d^2/10$ and $\tau_s=\rho_s d^2/60\eta$ give $I/\tau_s=6\eta\varphi$. ✓ By Newton's third law the friction torque density $\mathbf{A}\equiv(I/\tau_s)(\boldsymbol{\Theta}-\boldsymbol{\Omega})$ that the fluid exerts on the particles is exerted back on the fluid with sign $+$. A volume torque density $\mathbf{A}$ transmitted to the fluid is represented by an antisymmetric stress

$$\sigma^a_{ik}=-\tfrac{1}{2}\,\varepsilon_{ikl}A_l \quad\Rightarrow\quad f_i=\partial_k\sigma^a_{ik}=-\tfrac{1}{2}\varepsilon_{ikl}\partial_k A_l=+\tfrac{1}{2}(\nabla\times\mathbf{A})_i. \tag{28}$$

Why the factor $\frac{1}{2}$ and the sign are right (verification by conservation): the torque of the stress about any point is $\varepsilon_{ijk}x_j\partial_l\sigma_{kl}$; integrating the antisymmetric part over a volume yields exactly $\int A_i\,\mathrm{d}V$ delivered to orbital motion, so total (spin + orbital) angular momentum is conserved: $\mathrm{d}/\mathrm{d}t\int(\rho\,\mathbf{x}\times\mathbf{v}+I\boldsymbol{\Theta})\mathrm{d}V=\int\mathbf{M}\times\mathbf{H}\,\mathrm{d}V+$ surface terms, with the internal exchange $\mathbf{A}$ cancelling identically. ✓ Hence the momentum equation

$$\rho\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}=-\nabla p+(\mathbf{M}\cdot\nabla)\mathbf{H}+\eta\Delta\mathbf{v}+\frac{I}{2\tau_s}\,\nabla\times(\boldsymbol{\Theta}-\boldsymbol{\Omega}). \tag{SM 6}$$

6.3 Eliminating spin inertia ⇒ the PRL set

$|I\dot{\boldsymbol{\Theta}}|/|\frac{I}{\tau_s}(\boldsymbol{\Theta}-\boldsymbol{\Omega})|\sim\tau_s/\tau_B\sim10^{-5}$: drop the LHS of (SM 8):

$$\boxed{\boldsymbol{\Theta}=\boldsymbol{\Omega}+\frac{1}{6\eta\varphi}\mathbf{M}\times\mathbf{H}} \iff \boxed{6\eta\varphi(\boldsymbol{\omega}_p-\boldsymbol{\Omega})=\mathbf{M}\times\mathbf{H}} \tag{PRL 1}$$

[SM Eq. (11)]. Substituting into (SM 6) ($\frac{I}{2\tau_s}\cdot\frac{1}{6\eta\varphi}=\frac{1}{2}$):

$$\boxed{\rho\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}=-\nabla p+(\mathbf{M}\cdot\nabla)\mathbf{H}+\eta\Delta\mathbf{v}+\tfrac{1}{2}\nabla\times(\mathbf{M}\times\mathbf{H})} \tag{PRL 3, SM 12}$$

(the PRL misprints $\nabla\rho$ for ∇p), and into (SM 7), using $(\mathbf{M}\times\mathbf{H})\times\mathbf{M}=-\mathbf{M}\times(\mathbf{M}\times\mathbf{H})$:

$$\boxed{\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t}=\boldsymbol{\Omega}\times\mathbf{M}-\frac{1}{\tau_B}(\mathbf{M}-\mathbf{M}_0)-\frac{1}{6\eta\varphi}\mathbf{M}\times(\mathbf{M}\times\mathbf{H})} \tag{PRL 4, SM 13}$$

Note carefully: PRL (4) belongs to the *Debye/Shliomis-72 closure* branch. In the MRSh branch, (PRL 6) + (PRL 1) replace it, and the torque back-action is already inside (PRL 6).

7 The rotational-viscosity theorem: $\Delta\eta=\frac{3}{2}\eta\varphi\,(\Omega-\omega_p)/\Omega$

Suppose that, after averaging over the fast field oscillation (legitimate because the period $2\pi/\omega\ll\tau_h\sim\rho\ell^2/\eta$, the hydrodynamic time), the mean torque density is antiparallel to the local vorticity:

$$\overline{\mathbf{M}\times\mathbf{H}}=-6\eta\varphi\,g\,\boldsymbol{\Omega}. \tag{29}$$

Proof 1 (stress grouping). For incompressible flow $\Delta\mathbf{v} = -\nabla\times(\nabla\times\mathbf{v}) = -2\nabla\times\boldsymbol{\Omega}$, so

$$\eta\Delta\mathbf{v} + \tfrac{1}{2}\nabla\times\overline{\mathbf{M}\times\mathbf{H}} = -2\eta\nabla\times\boldsymbol{\Omega} - 3\eta\varphi g\,\nabla\times\boldsymbol{\Omega} = -2\eta\Big(1+\tfrac{3}{2}\varphi g\Big)\nabla\times\boldsymbol{\Omega}, \qquad \text{(PRL 10)}$$

i.e. Navier–Stokes with viscosity $\eta+\Delta\eta$,

$$\boxed{\Delta\eta = \tfrac{3}{2}\eta\varphi\, g}. \qquad (30)$$

By (PRL 1) and (29), $\boldsymbol{\omega}_p = (1-g)\boldsymbol{\Omega}$, hence

$$\boxed{\Delta\eta = \frac{3}{2}\,\eta\varphi\,\frac{\Omega-\omega_p}{\Omega}} \qquad \text{(PRL 2, SM 34)}$$

Proof 2 (SM's route, arbitrary amplitude). Average (PRL 1): $\overline{\mathbf{M}\times\mathbf{H}} = 6\eta\varphi(\overline{\boldsymbol{\Theta}}-\boldsymbol{\Omega})$; compare with (29): $g = (\Omega-\overline{\Theta})/\Omega$; insert into (30). This is exactly how SM promote their weak-field formula to all amplitudes [their Eqs. (37)–(39) and the sentence after]. Both proofs agree. ✓

Thermodynamic sign constraint (DC only). In a static field the field source does no work (a permanent magnet suffices). The mean dissipation must be ≥ 0 for the particle subsystem: the frictional power density is $\frac{I}{\tau_s}(\boldsymbol{\Theta}-\boldsymbol{\Omega})^2 = \zeta_r n(\boldsymbol{\omega}_p-\boldsymbol{\Omega})^2 \geq 0$, and stationarity of particle rotation requires the magnetic torque to be balanced by friction; one shows ω_p must lie *between* 0 and Ω, hence $g\in[0,1]$ and $0\leq\Delta\eta\leq\frac{3}{2}\eta\varphi$ for DC fields. An AC field is a power source; the bound disappears — negative g becomes possible, with the energy ledger: field → particle rotation → flow kinetic energy.

Part III
The three field regimes

8 Static field: McTague's experiment, Shliomis' formula, and the MRSh version

Geometry: $\mathbf{H} = H\mathbf{h}$ constant, $\mathbf{h}\perp\boldsymbol{\Omega}$ (in the PRL's cylindrical Poiseuille flow with axial field this holds at every point). Work to first order in $\Omega\tau_B \ll 1$.

8.1 Via the Shliomis-72 closure (the classical result)

Steady state of (SM 7) with $\mathbf{h}=\hat{\mathbf{x}}$, $\boldsymbol{\omega}_p=\omega_p\hat{\mathbf{z}}$, $\mathbf{M}=(M_x,M_y,0)$:

$$0 = -\omega_p M_y - \frac{M_x-M_0}{\tau_B}, \qquad 0 = \omega_p M_x - \frac{M_y}{\tau_B} \;\Rightarrow\; M_y = \omega_p\tau_B M_x, \quad M_x = \frac{M_0}{1+\omega_p^2\tau_B^2} \simeq M_0. \qquad (31)$$

Torque: $(\mathbf{M}\times\mathbf{H})_z = -M_y H = -\omega_p\tau_B M_0 H$. Insert into (PRL 1) and use

$$\tau_B M_0 H = \frac{3\eta V}{kT}\,\frac{\varphi}{V}\,mL(\xi)H = 3\eta\varphi\,\xi L(\xi): \qquad 6\eta\varphi(\omega_p-\Omega) = -3\eta\varphi\,\xi L\,\omega_p \;\Rightarrow\; \omega_p = \frac{\Omega}{1+\frac{1}{2}\xi L}. \qquad (32)$$

Then (PRL 2, SM 34):

$$\Delta\eta = \frac{3}{2}\eta\varphi\,\frac{\xi L}{2+\xi L} = \frac{3}{2}\eta\varphi\,\frac{\xi\coth\xi-1}{\xi\coth\xi+1} \;\Rightarrow\; \boxed{\Delta\eta = \frac{3}{2}\,\eta\varphi\,\frac{\xi-\tanh\xi}{\xi+\tanh\xi}\,\sin^2\beta} \qquad \text{(SM 2)}$$

($\sin^2\beta$ restores a general angle β between $\mathbf{H}$ and $\boldsymbol{\Omega}$: only the perpendicular field component produces torque; here $\beta=\pi/2$). Limits [SM Eqs. (3),(4)]: $\Delta\eta = \frac{1}{4}\eta\varphi\xi^2$ ($\xi\ll 1$), $\Delta\eta\to\frac{3}{2}\eta\varphi$ ($\xi\to\infty$: particles fully locked, $\omega_p=0$, $g=1$). This is the theory that matches McTague's 1969 rotating-field-free measurements and the $f=0$ curve of PRL Fig. 2.

8.2 Via the MRSh closure

Now do the same with (PRL 6). First order in $\Omega\tau_B$: $\boldsymbol{\xi}_e = \xi\mathbf{h} + \boldsymbol{\delta}$, $\boldsymbol{\delta} \perp \mathbf{h}$. Longitudinal balance gives $\xi_e = \xi$ unchanged. Transverse balance: using $\boldsymbol{\xi}_e \times (\boldsymbol{\xi}_e \times \boldsymbol{\xi})\big|_\perp = \xi^2\boldsymbol{\delta}$ (computed as in Sec. 5.3),

$$0 = L\,\boldsymbol{\Omega}\times\mathbf{h} - \Big[\frac{L}{\tau_B\xi} + \frac{1}{2\tau_B}\Big(1 - \frac{3L}{\xi}\Big)\Big]\boldsymbol{\delta} = L\,\boldsymbol{\Omega}\times\mathbf{h} - \frac{\xi - L}{2\tau_B\,\xi}\,\boldsymbol{\delta}, \tag{33}$$

$$\boldsymbol{\delta} = \frac{2\tau_B\,\xi L}{\xi - L}\,\boldsymbol{\Omega}\times\mathbf{h}, \qquad \mathbf{M}_\perp = \varphi M_s\frac{L}{\xi}\boldsymbol{\delta} = \varphi M_s\,\frac{2\tau_B L^2}{\xi - L}\,\boldsymbol{\Omega}\times\mathbf{h}. \tag{34}$$

Torque: $\mathbf{M}_\perp \times \mathbf{H} = \varphi M_s H\,\frac{2\tau_B L^2}{\xi - L}(\boldsymbol{\Omega}\times\mathbf{h})\times\mathbf{h} = -\varphi M_s H\tau_B\,\frac{2L^2}{\xi - L}\,\boldsymbol{\Omega}$ (using $(\boldsymbol{\Omega}\times\mathbf{h})\times\mathbf{h} = -\boldsymbol{\Omega}$ for $\boldsymbol{\Omega} \perp \mathbf{h}$), and $\varphi M_s H\tau_B = 3\eta\varphi\xi$:

$$\overline{\mathbf{M}\times\mathbf{H}} = -6\eta\varphi\,\frac{\xi L^2}{\xi - L}\,\boldsymbol{\Omega} \;\Rightarrow\; \boxed{\Delta\eta^{\mathrm{MRSh}} = \frac{3}{2}\eta\varphi\,\frac{\xi L^2(\xi)}{\xi - L(\xi)}}. \tag{35}$$

Checks: $\xi \to 0$: $L \simeq \xi/3$, $\xi - L \simeq 2\xi/3$, ratio $\to \xi^2/6 \cdot \frac{3}{2}\eta\varphi = \frac{1}{4}\eta\varphi\xi^2$ — agrees with (SM 2) ✓. $\xi \to \infty$: ratio $\to 1$, saturation $\frac{3}{2}\eta\varphi$ — agrees ✓.

8.3 The two closures are *not* identical at intermediate fields

Numerical comparison of $g(\xi)$ (computed to 4 digits):

ξ	0.5	1	2	3	5	10
$g^{\mathrm{MRSh}} = \xi L^2/(\xi - L)$	0.0400	0.1426	0.3948	0.5812	0.7621	0.8901
$g^{\mathrm{Sh72}} = (\xi - \tanh\xi)/(\xi + \tanh\xi)$	0.0394	0.1353	0.3495	0.5019	0.6667	0.8182

Up to 16% apart around $\xi \sim 3$–5, converging at both ends. MRSh (being the better closure of the FP hierarchy) is the more accurate; at $\varphi = 0.2$ with strong aggregation the experiment cannot resolve the difference, and the PRL sidesteps it by normalizing to the *measured* $\eta_r(\infty, 0)$. Be aware of this when doing quantitative follow-up work.

8.4 Confrontation with the experiment's absolute numbers (critical)

Dilute theory: $\eta_r(\infty, 0) = \frac{3}{2}\varphi = 0.30$. Experiment: $\eta(0) = 77$ cP $\to \eta(2200\,\mathrm{Oe}) = 220$ cP, i.e. $\eta_r(\infty, 0) \simeq 2.5$ — a factor ~ 8. The single-particle theory therefore predicts the correct *shape* of $g(\xi, \omega\tau_B)$ but not the magnitude at $\varphi = 0.2$; interparticle correlations/chains renormalize the amplitude. The PRL's Figs. 3–5 comparison is a shape comparison after empirical normalization.

9 Weak alternating field (SM Sec. II): the sign change from first principles

Field $\mathbf{H} = H_0\cos\omega t\,\hat{\mathbf{x}}$; weak: $\xi_0 = mH_0/kT \ll 1$, so the Debye equation (20) is *exact* here (Sec. 5) — this regime is closure-independent. Vorticity $\boldsymbol{\Omega} = \Omega\hat{\mathbf{z}}$, $\Omega\tau_B \ll 1$.

9.1 Zeroth order (quiescent fluid)

x-component of (20): $\dot{M}_x = -(M_x - \chi H_0\cos\omega t)/\tau_B$. Periodic solution by phasors, $M_x = \mathrm{Re}[\chi H_0 e^{i\omega t}/(1 + i\omega\tau_B)]$:

$$M_x = \chi H_0\cos\alpha\,\cos(\omega t - \alpha), \qquad \tan\alpha = \omega\tau_B. \tag{SM 30}$$

$M_y = 0$: no torque, no particle spin. *Physical reading (SM's "first variant"):* the linear field is the sum of two counter-rotating circular fields; in a quiescent fluid the two half-populations of particles swing clockwise/counterclockwise symmetrically; $\langle\boldsymbol{\Theta}\rangle = 0$ by symmetry.

9.2 First order in $\Omega\tau_B$: vorticity breaks the degeneracy

y-component of (20): $\dot{M}_y = \Omega M_x - M_y/\tau_B$ (the equilibrium term has no y part; the torque term of PRL (4) is $O(\xi_0^3)$, negligible). Phasors: with $\tilde{M}_x = \chi H_0 \cos\alpha\, e^{-i\alpha}$,

$$\tilde{M}_y = \frac{\Omega\tau_B\, \tilde{M}_x}{1 + i\omega\tau_B} = \Omega\tau_B\chi H_0 \cos^2\alpha\, e^{-2i\alpha} \;\Rightarrow\; \boxed{M_y = \Omega\tau_B\chi H_0 \cos^2\alpha\, \cos(\omega t - 2\alpha)} \qquad \text{(SM 23, 32)}$$

Equivalently (SM's first variant): the co-rotating circular component lags by $(\omega - \Omega)\tau_B$, the counter-rotating one by $(\omega + \Omega)\tau_B$ [SM (21)]; their superposition no longer cancels in y. Both routes give the same M_y — SM verify this explicitly, and so did we.

9.3 Mean torque and $\Delta\eta(\omega)$; the amplitude/rms bookkeeping

$$(\mathbf{M}\times\mathbf{H})_z = -M_y H_x = -\Omega\tau_B\chi H_0^2 \cos^2\alpha\, \cos(\omega t - 2\alpha)\cos\omega t, \qquad (36)$$

$$\overline{(\mathbf{M}\times\mathbf{H})_z} = -\tfrac{1}{2}\Omega\tau_B\chi H_0^2 \cos^2\alpha\, \cos 2\alpha, \qquad \tau_B\chi H_0^2 = \frac{3\eta V}{kT}\frac{nm^2}{3kT}H_0^2 = \eta\varphi\,\xi_0^2. \qquad (37)$$

Comparing with (29) and using (30):

$$g = \frac{\xi_0^2}{12}\cos^2\alpha\cos 2\alpha, \qquad \boxed{\Delta\eta = \frac{1}{8}\,\eta\varphi\,\xi_0^2\,\frac{1-\omega^2\tau_B^2}{(1+\omega^2\tau_B^2)^2} = \frac{1}{4}\,\eta\varphi\,\bar{\xi}^{\,2}\,\frac{1-\omega^2\tau_B^2}{(1+\omega^2\tau_B^2)^2}} \qquad \text{(SM 28)}$$

with $\bar{\xi} = \xi_0/\sqrt{2}$ the *rms* Langevin parameter. SM write the prefactor $\frac{1}{4}$ because they define ξ over the acting field $H_0/\sqrt{2}$ (stated below their Eq. (18)); with the amplitude convention it is $\frac{1}{8}$. *Consistency check of the factor:* at $\omega \to 0$ the field is quasistatic and $\Delta\eta$ must equal the time average of the instantaneous static value $\frac{1}{4}\eta\varphi\xi_0^2\cos^2\omega t$, i.e. $\frac{1}{8}\eta\varphi\xi_0^2$ — which (SM 28) gives. ✓ (This check also exposed a factor-2 slip in an earlier version of these notes; see Sec. 16.) A second, independent check: direct numerical integration of the full MRSh system at $\xi_0 = 0.3$ reproduces $g = \xi_0^2 R(\omega\tau_B)/12$ to $< 1\%$ (Sec. 12).

9.4 All qualitative results, verified

Let $x = \omega\tau_B$, $R(x) = (1-x^2)/(1+x^2)^2$:

1. Sign change at $x = 1$: $\omega_0 = \tau_B^{-1}$, independent of amplitude (in this order).
2. Extremum: $R'(x) = \dfrac{2x(x^2-3)}{(1+x^2)^3} = 0 \Rightarrow x = \sqrt{3}$; $R(\sqrt{3}) = -\frac{1}{8}$; so $\Delta\eta_{\min} = -\Delta\eta(0)/8$ at $\omega_m\tau_B = \sqrt{3} \approx 1.732$. (Numerically confirmed: $R(\sqrt{3}) = -0.125$.)
3. High frequency: $R \simeq -x^{-2}$, and with $\bar{\xi}/\tau_B = M_s H_0/(3\sqrt{2}\,\eta)$,

$$\Delta\eta \simeq -\frac{\eta\varphi\bar{\xi}^2}{4\omega^2\tau_B^2} = -\frac{\varphi}{\eta}\Big(\frac{M_s\bar{H}}{6\omega}\Big)^2, \qquad \bar{H} = H_0/\sqrt{2}, \qquad \text{(SM 29)}$$

   so $\Delta\eta \to 0^-$: a too-fast field is invisible to the particles — but the approach to zero is from *below*, overturning the pre-1994 expectation of a monotone decay of $\Delta\eta(\omega)$ from its static value.
4. Mean particle spin [SM (33)]: averaging (PRL 1) with the torque above, $\Theta = \Omega\big[1 - \frac{\bar{\xi}^2}{6}R(\omega\tau_B)\big]$; $\omega\tau_B < 1 \Rightarrow \Theta < \Omega$ (drag, $\Delta\eta > 0$); $\omega\tau_B > 1 \Rightarrow \Theta > \Omega$ (spin-up, $\Delta\eta < 0$). Substituting into (PRL 2, SM 34) reproduces (SM 28) identically. ✓

Mechanism in one sentence: vorticity biases the field-driven rotational swings; below $\omega\tau_B = 1$ the field mainly *orients* (hindrance), above it the retarded response phase-locks so that the transverse magnetization torques the particles *forward*, converting field energy into flow kinetic energy.

10 Arbitrary amplitude with the Debye closure (SM Sec. III), including a corrected Eq. (40)

10.1 Reduction to two scalar ODEs [SM (35)→(36)]

Keep the Shliomis-72 closure, PRL (4)/SM (13), with $\mathbf{M}_0 = nmL[\xi(t)]\mathbf{h}$, $\xi(t) = \xi\gamma(t)$, $|\gamma| \le 1$ periodic. In the quiescent-fluid zeroth order plus first order in $\Omega\tau_B$, write

$$M_x = nm\,F(t), \qquad M_y = nm\,\Omega\tau_B\,G(t). \tag{38}$$

Project SM (13). The x-equation at zeroth order:

$$\tau_B \frac{\mathrm{d}F}{\mathrm{d}t} = L[\xi\gamma(t)] - F. \tag{SM 36a}$$

The y-equation at first order: convection gives ΩM_x; relaxation gives $-M_y/\tau_B$; the nonlinear torque term of SM (13) contributes, per the algebra of the triple product with $\mathbf{M}$ nearly along $\mathbf{h}$, $-\frac{\tau_s}{I}M^2 H\,M_y \to -\frac{1}{2}\xi\gamma F\,G/\tau_B$ in these units, yielding

$$\tau_B \frac{\mathrm{d}G}{\mathrm{d}t} = F - \Big(1 + \tfrac{1}{2}\,\xi\gamma(t)F(t)\Big)G. \tag{SM 36b}$$

Check: weak field, harmonic $\gamma = \cos\omega t$: drop the nonlinear factor, $F \to \frac{\xi}{3}\cos\alpha\cos(\omega t - \alpha)$, $G \to \frac{\xi}{3}\cos^2\alpha\cos(\omega t - 2\alpha)$ — exactly M_x, M_y of Sec. 9. ✓

10.2 Torque, viscosity, and the general proof of PRL Eq. (2) [SM (37)–(39)]

$$\overline{(\mathbf{M}\times\mathbf{H})_z} = -\overline{M_y H_x} = -nmH\,\Omega\tau_B\,\overline{G\gamma} = -3\eta\varphi\,\xi\,\Omega\,\overline{G\gamma} \tag{SM 37}$$

(using $nmH\tau_B = 3\eta\varphi\xi$ as before). Comparing with (29): $g = \frac{1}{2}\xi\overline{G\gamma}$ and

$$\Delta\eta = \tfrac{3}{4}\eta\varphi\,\xi\,\overline{G\gamma}. \tag{SM 38}$$

Averaging (PRL 1) directly: $\overline{\mathbf{M}\times\mathbf{H}} = -6\eta\varphi(\mathbf{\Omega} - \overline{\mathbf{\Theta}})$; equate with SM (37):

$$\tfrac{1}{2}\xi\,\overline{G\gamma} = \frac{\Omega - \overline{\Theta}}{\Omega} \;\Rightarrow\; \Delta\eta = \tfrac{3}{2}\eta\varphi\,\frac{\Omega - \overline{\Theta}}{\Omega}, \tag{39}$$

i.e. the weak-field formula (PRL 2, SM 34) is *exact at arbitrary amplitude and frequency* within this closure. This is SM's key structural theorem, and the PRL uses it verbatim.

10.3 Square-wave field: full derivation of the analytic weak-field result

Let $\gamma(t)$ be the ±1 square wave of period $T = 2\pi/\omega$. Weak field: linearize (36):

$$\tau_B\dot{F} = \tfrac{\xi}{3}\gamma(t) - F, \qquad \tau_B\dot{G} = F - G. \tag{40}$$

Both equations are antiperiodic-compatible: since $\gamma(t + T/2) = -\gamma(t)$ and the system is linear, the attractor obeys $F(t + T/2) = -F(t)$, $G(t + T/2) = -G(t)$ [SM state $G(t) = -G(t + \pi/\omega)$]. Work on $t \in (0, T/2)$ where $\gamma = +1$; define

$$\varepsilon \equiv \frac{T/4}{\tau_B} = \frac{\pi}{2\omega\tau_B}, \qquad E \equiv e^{-2\varepsilon} = e^{-(T/2)/\tau_B}. \tag{41}$$

Step 1: F. $F(t) = \frac{\xi}{3} + (F_0 - \frac{\xi}{3})e^{-t/\tau_B}$ with $F_0 = F(0)$. Antiperiodicity $F(T/2) = -F_0$:

$$\tfrac{\xi}{3} + (F_0 - \tfrac{\xi}{3})E = -F_0 \;\Rightarrow\; F_0 = -\tfrac{\xi}{3}\,\frac{1-E}{1+E} = -\tfrac{\xi}{3}\tanh\varepsilon, \qquad F(t) = \tfrac{\xi}{3}\Big[1 - (1+\tanh\varepsilon)e^{-t/\tau_B}\Big]. \tag{42}$$

Step 2: G. With $F = a + be^{-t/\tau_B}$, $a = \frac{\xi}{3}$, $b = -a(1+\tanh\varepsilon)$, the resonant particular solution is $a + (bt/\tau_B)e^{-t/\tau_B}$, so $G(t) = a + (bt/\tau_B)e^{-t/\tau_B} + Ce^{-t/\tau_B}$. Antiperiodicity $G(T/2) = -G(0)$:

$$C = -\frac{2a + 2\varepsilon bE}{1+E}. \tag{43}$$

Step 3: the average. By antiperiodicity $\overline{G\gamma} = \frac{2}{T}\int_0^{T/2} G\,\mathrm{d}t$. Using $\int_0^{2\varepsilon\tau_B}(bt/\tau_B)e^{-t/\tau_B}\mathrm{d}t = b\tau_B[1-(1+2\varepsilon)E]$ and $\int_0^{2\varepsilon\tau_B} Ce^{-t/\tau_B}\mathrm{d}t = C\tau_B(1-E)$:

$$\frac{\overline{G\gamma}}{a} = 1 + \frac{1}{2\varepsilon}\Big\{\frac{b}{a}\big[1-(1+2\varepsilon)E\big] + \frac{C}{a}(1-E)\Big\}. \tag{44}$$

Insert $b/a = -2/(1+E)$, $C/a = -2/(1+E) + 4\varepsilon E/(1+E)^2$; the braces collapse (three lines of algebra, using $\tanh\varepsilon = \frac{1-E}{1+E}$ and $\operatorname{sech}^2\varepsilon = \frac{4E}{(1+E)^2}$) to $-4\tanh\varepsilon + 2\varepsilon\operatorname{sech}^2\varepsilon$, giving

$$\overline{G\gamma} = \tfrac{\xi}{3}\Big[1 + \operatorname{sech}^2\varepsilon - \tfrac{2\tanh\varepsilon}{\varepsilon}\Big] = \tfrac{\xi}{3}\Big[2 - \tanh^2\varepsilon - \tfrac{2\tanh\varepsilon}{\varepsilon}\Big], \tag{45}$$

and finally, via SM (38),

$$\boxed{\Delta\eta = \frac{1}{4}\eta\varphi\xi^2\Big(2 - \tanh^2\varepsilon - \frac{2\tanh\varepsilon}{\varepsilon}\Big), \qquad \varepsilon = \frac{\pi}{2\omega\tau_B}} \qquad \text{(SM 40, corrected)}$$

Checks: $\varepsilon \to \infty$ (static): $2 - 1 - 0 = 1 \Rightarrow \Delta\eta = \frac{1}{4}\eta\varphi\xi^2$, the static weak-field value ($\gamma^2 \equiv 1$ for a square wave, so no $\frac{1}{2}$ here) ✓. $\varepsilon \to 0$ (high ω): $2 - \varepsilon^2 - 2(1-\varepsilon^2/3) \to -\varepsilon^2/3 < 0$ ✓. Numerically: zero of (SM 40, corrected) at $\varepsilon_0 = 1.6061 \Rightarrow \omega_0\tau_B = \pi/2\varepsilon_0 = 0.978$; minimum at $\varepsilon_m = 0.9199 \Rightarrow \omega_m\tau_B = 1.7075$, depth $-0.1050\,\Delta\eta(0)$ — reproducing SM's quoted $\omega_0\tau_B = 0.98$, $\omega_m\tau_B = 1.707$, $\Delta\eta_m = -0.105\,\Delta\eta(0)$ to all quoted digits. ✓✓✓

Note (OCR/typo). The scanned SM paper appears to read "$2 - \tanh\varepsilon - 2\tanh\varepsilon/\varepsilon$". That expression fails all three of SM's own quoted numbers (it would give $\omega_0\tau_B = 0.87$, $\omega_m\tau_B = 2.29$, depth -0.33). The form (SM 40, corrected) with $\tanh^2\varepsilon$ reproduces them exactly, and is what our independent derivation yields. Conclusion: the printed/scanned formula dropped a square; use (SM 40, corrected).

10.4 Where this closure fails — and why the PRL needed MRSh

Solving SM (36) at finite ξ (their Fig. 3–5) predicts: the $\Delta\eta(\xi)$ curves have their maxima at $\omega\tau_B < 1$ and show *no* minimum for $\omega\tau_B > 1$; in strong fields $\Delta\eta < 0$ at essentially all frequencies. The experiment (PRL Fig. 2) shows the *inverse*: pronounced negative minima develop at $\omega\tau_B > 1$ and deepen with frequency. The origin of the failure is Sec. 5: at finite ξ the true transverse/longitudinal relaxation times shorten ($\tau_\perp \simeq 2\tau_B/\xi$, $\tau_\parallel \simeq \tau_B/\xi$), while the Debye closure keeps a rigid τ_B — so it misplaces the frequency scale as ξ grows. The MRSh closure carries the field-dependent times automatically.

Part IV
The PRL's theory, numerics, and experiment

11 Perturbation theory in $\Omega\tau_B$ for the MRSh equation: PRL Eqs. (7)–(9)

Small parameter: transient birefringence gives $\tau_B \simeq 1.6$ ms and $\Omega \lesssim 60\ \mathrm{s}^{-1}$ in the capillary, so $\Omega\tau_B \lesssim 0.1$. Expand

$$\boldsymbol{\xi}_e = \xi_e(t)\mathbf{h} + \boldsymbol{\delta}(t), \qquad \boldsymbol{\delta} \perp \mathbf{h} = O(\Omega\tau_B), \qquad \boldsymbol{\xi} = \xi_0\cos\omega t\,\mathbf{h}. \tag{46}$$

11.1 Zeroth order

Project (PRL 6) on $\mathbf{h}$ (the triple-product term has no longitudinal first-order part): $L'(\xi_e)\dot{\xi}_e = -\frac{L_e}{\tau_B \xi_e}(\xi_e - \xi_0 \cos\omega t)$, i.e.

$$\boxed{\tau_B \frac{\mathrm{d}\xi_e}{\mathrm{d}t} = -\Big(\frac{\mathrm{d}\ln L_e}{\mathrm{d}\xi_e}\Big)^{-1}\Big(1 - \frac{\xi_0}{\xi_e}\cos\omega t\Big)}, \qquad \Big(\frac{\mathrm{d}\ln L_e}{\mathrm{d}\xi_e}\Big)^{-1} = \frac{L_e}{L'(\xi_e)}. \tag{PRL 7}$$

Nonlinear magnitude relaxation with the instantaneous time $\tau_\parallel(\xi_e)$ of Sec. 5. *Regularity check:* as $\xi_e \to 0$, $L_e/L' \to \xi_e$, so the RHS $\to -(\xi_e - \xi_0\cos\omega t)$: plain Debye, no singularity; ξ_e should be treated as a *signed* variable that passes smoothly through zero each half-period (all functions involved are odd/even consistently).

11.2 First order: derivation of PRL Eq. (8), line by line

Ansatz for the transverse magnetization,

$$\mathbf{M}^{(1)} = M^{(0)} F(t)\,\tau_B\,\boldsymbol{\Omega}\times\mathbf{h}, \qquad M^{(0)} = \varphi M_s L(\xi_e), \tag{47}$$

equivalently the transverse part of $L_e\hat{\boldsymbol{\xi}}_e$ is $\mathbf{u} = L_e F \tau_B\,\boldsymbol{\Omega}\times\mathbf{h}$ and $\boldsymbol{\delta} = \xi_e F\tau_B\,\boldsymbol{\Omega}\times\mathbf{h}$. Transverse projection of (PRL 6) at $O(\Omega\tau_B)$: (a) convection $\to L_e\,\boldsymbol{\Omega}\times\mathbf{h}$; (b) longitudinal-relaxation term $\to -\frac{L_e}{\tau_B\xi_e}\boldsymbol{\delta}$; (c) triple product $\to -\frac{1}{2\tau_B}\big(1-\frac{3L_e}{\xi_e}\big)\frac{\xi_0\cos\omega t}{\xi_e}\boldsymbol{\delta}$ (transverse part $= \xi_e\,\xi_0\cos\omega t\,\boldsymbol{\delta}$, computed exactly as in Sec. 5.3). Hence

$$\tau_B\frac{\mathrm{d}(L_eF)}{\mathrm{d}t} = L_e - L_eF - \frac{F}{2}\Big(1-\frac{3L_e}{\xi_e}\Big)\xi_0\cos\omega t. \tag{48}$$

Now use the zeroth-order result in the form $\tau_B\dot{L}_e = -L_e\big(1-\frac{\xi_0\cos\omega t}{\xi_e}\big)$ (multiply (PRL 7) by L'), expand the LHS, divide by L_e:

$$-F\Big(1-\frac{\xi_0\cos\omega t}{\xi_e}\Big) + \tau_B\dot{F} = 1 - F - \frac{F}{2}\Big(\frac{1}{L_e}-\frac{3}{\xi_e}\Big)\xi_0\cos\omega t, \tag{49}$$

$$\tau_B\dot{F} = 1 - \frac{F\xi_0\cos\omega t}{2}\Big[\frac{2}{\xi_e}+\frac{1}{L_e}-\frac{3}{\xi_e}\Big] \;\Rightarrow\; \boxed{\tau_B\frac{\mathrm{d}F}{\mathrm{d}t} = 1 - \frac{1}{2}\Big(\frac{1}{L_e}-\frac{1}{\xi_e}\Big)\xi_0 F\cos\omega t} \tag{PRL 8}$$

Checks: (i) weak field: $\frac{1}{L_e}-\frac{1}{\xi_e}\to\frac{2}{\xi_e}$; solving (PRL 7)–(PRL 8) then reproduces Sec. 9 exactly (verified numerically below). (ii) Static field: $\dot{F}=0 \Rightarrow F = \big[\frac{1}{2}\big(\frac{1}{L}-\frac{1}{\xi}\big)\xi\big]^{-1} = \dfrac{2L}{\xi - L}$; then $g = \frac{1}{2}\xi L F = \dfrac{\xi L^2}{\xi - L}$, reproducing the MRSh static formula (35) exactly. ✓

11.3 Mean torque: PRL Eq. (9)

$\mathbf{M}^{(0)} \parallel \mathbf{H}$: torqueless. With $(\boldsymbol{\Omega}\times\mathbf{h})\times\mathbf{h} = -\boldsymbol{\Omega}$ (valid since $\boldsymbol{\Omega}\perp\mathbf{h}$ in the tube):

$$\mathbf{M}\times\mathbf{H} = \mathbf{M}^{(1)}\times\mathbf{H} = -\varphi M_s L(\xi_e) F\,\tau_B H_0\cos\omega t\,\boldsymbol{\Omega} = -3\eta\varphi\,\xi_0\cos\omega t\,L(\xi_e)F\,\boldsymbol{\Omega}, \tag{50}$$

using $\varphi M_s H_0 \tau_B = 3\eta\varphi\xi_0$ once more. Averaging over one period of the attractor:

$$\boxed{\overline{\mathbf{M}\times\mathbf{H}} = -6\eta\varphi\,g\,\boldsymbol{\Omega}, \qquad g = \tfrac{1}{2}\,\xi_0\,\overline{\cos\omega t\;L(\xi_e)F}} \tag{PRL 9}$$

and by the theorem of Sec. 7: $\Delta\eta = \frac{3}{2}\eta\varphi\,g$, $\boldsymbol{\omega}_p = (1-g)\boldsymbol{\Omega}$ [PRL (2),(10)]. The whole problem is now two scalar ODEs and one quadrature.

11.4 A numerically regular formulation (practical note)

Equation (PRL 8) contains $1/L_e$, which diverges when ξ_e crosses zero. The physical transverse moment is $P \equiv L(\xi_e)F$ (it is P, not F, that enters the torque), and its equation is regular:

$$\tau_B \frac{\mathrm{d}P}{\mathrm{d}t} = L(\xi_e) - P - \frac{P}{2}\, s(\xi_e)\, \xi_0 \cos\omega t, \qquad s(\xi) \equiv \frac{1}{L(\xi)} - \frac{3}{\xi} = \frac{\xi}{5} + O(\xi^3), \tag{51}$$

with s a smooth odd function. Integrate (ξ_e, P) by RK4 (as the PRL does, from $\xi_e(0) = \xi_0$, $P(0) \approx 0$), then $g = \frac{1}{2}\xi_0 \overline{\cos\omega t\, P}$.

12 Numerical verification of the PRL's theoretical curves

We integrated the (ξ_e, P) system (RK4, ≥ 1200 steps/period, ≥ 20 periods to reach the attractor) and computed g:

$g(\xi_0, \omega\tau_B)$	ξ_0=4	8	12	16	20
$\omega\tau_B = 0.5$	0.295	0.451	0.517	0.554	0.580
$\omega\tau_B = 1.0$	0.047	0.163	0.231	0.275	0.306
$\omega\tau_B = 1.5$	−0.105	−0.110	−0.044	0.009	0.049
$\omega\tau_B = 2.0$	−0.140	−0.312	−0.299	−0.246	−0.197

Compare with PRL Fig. 3: the $\omega\tau_B = 0.5$ curve rising to ~ 0.6 by $\xi_0 = 20$; $\omega\tau_B = 1$ small and positive; $\omega\tau_B = 1.5$ dipping to ≈ -0.11 near $\xi_0 \sim 6$–8 and re-crossing zero near $\xi_0 \sim 15$; $\omega\tau_B = 2$ with a minimum ≈ -0.31 near $\xi_0 \sim 9$ and slow nonmonotonic recovery — all in quantitative agreement with the published curves. Weak-field check: at $\xi_0 = 0.3$ the same code returns $g = \xi_0^2 R(\omega\tau_B)/12$ within 1% at $\omega\tau_B = 0.3, 1, \sqrt{3}, 3$ (including the exact zero at $\omega\tau_B = 1$), confirming both the code and the amplitude-convention factor in (SM 28).

Isolines of g in the $(\xi_0, \omega\tau_B)$ plane (PRL Fig. 4): the separatrix $g = 0$ is nearly vertical at $\omega\tau_B \approx 1$ for moderate ξ_0 (the weak-field criterion survives), while $|g|$ grows with ξ_0 toward $O(1)$ saturation — positive branch to the left, negative to the right, most negative isoline -0.9 at large $\xi_0, \omega\tau_B$.

13 The experiment, its working equations, and quantitative checks

13.1 Viscometry

Horizontal capillary, $d_c = 1$ mm, coaxial inside a solenoid: $\mathbf{H} \parallel$ flow $\Rightarrow \mathbf{H} \perp \boldsymbol{\Omega}$ pointwise, $\beta = \pi/2$ (maximal effect, and no Kelvin force). Hydrostatic head h drives Poiseuille flow; for a Newtonian fluid the mass flow rate obeys

$$Q = \frac{\pi \rho^2 g_{\mathrm{grav}}\, h\, d_c^4}{128\, \eta\, \ell} \;\propto\; \frac{h}{\eta}. \tag{52}$$

PRL Fig. 1: $Q(h)$ linear at $H = 0$ ($\eta = 77$ cP) and at $H = 2200$ Oe static ($\eta = 220$ cP): Newtonian in both states; slopes give η. Sample: Massart-method Co-ferrite ferrofluid in water, $\varphi = 0.2$, $d \approx 10$ nm.

13.2 Data reduction and conversions (re-derived in Sec. 2)

$$\eta_r(H, f) = \frac{\eta(H, f) - \eta(0, 0)}{\eta(0, 0)}, \qquad \xi_0 = 5.25 \times 10^{-3} H(\mathrm{Oe}), \qquad \omega\tau_B = 10^{-2} f(\mathrm{Hz}), \tag{53}$$

the latter equivalent to $\tau_B = 1.59$ ms, matching the birefringence value. Our recomputation of ξ_0/H gives 5.1×10^{-3} at $T = 295$ K (5.25×10^{-3} corresponds to $T \approx 289$ K): consistent.

13.3 Findings vs. theory

- $f = 0$: monotone McTague curve, $\eta_r \to 2.5$.
- $f = 52$ Hz ($\omega\tau_B = 0.52$): same shape, smaller (positive-g branch).
- $f = 130$ Hz ($\omega\tau_B = 1.3$): η_r decreases, turns *negative*, minimum, then rises — the predicted sign change at $\omega\tau_B \sim 1$ with the finite-amplitude nonmonotonicity of Fig. 3.
- $f = 345$, 645, 1480 Hz: deeper minima with frequency, as in the $\omega\tau_B = 1.5$–2 theory rows of Sec. 12. (At 1480 Hz the supply limits $H < 800$ Oe.)
- Fixed $H \sim 2000$ Oe ($\xi_0 = 10.5$): $\eta = 220, 100, 77, 58$ cP at $f = 0, 50, 250, 700$ Hz, i.e. $\Delta\eta$ from $+143$ cP to -19 cP; total viscosity 25% *below* the zero-field value ($58/77 = 0.75$).
- **PRL Fig. 5**: experimental isolines of $g = \eta_r/2.5$, from $+0.216$ to -0.092, topologically matching theoretical Fig. 4, with the $g = 0$ line at $\omega\tau_B \sim 1$ in both.

Consistency check of the deepest measured value: $\Delta\eta = -19$ cP $\Rightarrow g = \eta_r/2.5 = (-19/77)/2.5 = -0.099$, i.e. the outermost experimental isoline -0.092 region — and our computed $g(\xi_0 \simeq 10.5, \omega\tau_B = 7)$ from the model sits at the -0.1 level after the same normalization. Coherent.

13.4 Critical assessment (facts the paper does not emphasize)

1. τ_B **inconsistency.** Single 10-nm grain in water: $\tau_B = 4 \times 10^{-7}$ s; in the suspension ($\eta = 77$ cP): 3×10^{-5} s. Measured: 1.6×10^{-3} s. Inverting (3) gives hydrodynamic diameter ~ 160 nm (with solvent viscosity): the rotating units are aggregates/chains. The theory is applied with the *measured* τ_B but the *single-grain* m in ξ_0 — a hybrid parametrization.
2. **Amplitude anomaly.** $\eta_r(\infty, 0) = 2.5$ vs. dilute $\frac{3}{2}\varphi = 0.3$ (factor 8): shape-only agreement, empirical normalization.
3. **Closure ambiguity** at intermediate ξ (Sec. 8): up to 16% between Sh-72 and MRSh static curves — below the experiment's resolving power here, but relevant for precision follow-ups.

14 Zero total viscosity and the "pump" instability (PRL conclusion; SM Sec. IV)

$\eta_{\text{total}} = \eta + \Delta\eta = 0$ requires $\Delta\eta = -\eta$, i.e.

$$\frac{\Delta\eta(\xi, \omega\tau_B)}{\Delta\eta(\infty, 0)} = -\frac{\eta}{\frac{3}{2}\eta\varphi} = -\frac{2}{3\varphi} \overset{\varphi=0.2}{=} -\frac{10}{3}, \qquad \text{i.e.} \qquad g = -\frac{2}{3\varphi}. \tag{54}$$

Equivalent stability statement: consider a ferrofluid at rest between rigid walls $x = 0, \ell$ under $H_x = H_0 \cos\omega t$; perturb with $v = w(t) \sin qx\, \hat{\mathbf{y}}$, $M_y = f(t) \cos qx$. Linearizing SM (12)–(13) gives an effective damping $\propto (\eta + \Delta\eta)q^2$; boundary conditions quantize $q = \pi r/\ell$; the least damped mode is $q = \pi/\ell$, and it grows exactly when $\eta_{\text{total}} < 0$ — states of negative total viscosity are *unstable states of rest* (spontaneous recirculation; nonzero flux through the cross-section: "the system works as a pump"), not a steady material property. On the critical isoline SM find $\xi_c = 126$, $\omega_c\tau_B = 20$: with the present particles ξ_c demands $H \sim 2.4 \times 10^4$ Oe $> H_a$ (rigid-dipole model breaks), and larger particles ruin stability/Newtonian behavior — so unattainable here. What is attained: viscosity tuned continuously 220 $\to$ 50 cP at fixed $H = 2000$ Oe by sweeping $f = 0 \to 700$ Hz ("adaptive dampers").

15 Verification ledger

Result	Verification performed
$\tau_B = 3\eta V/kT$	Einstein relation + $l(l+1)D_r$; rate re-derived from FP diffusion term (Sec. 4)
$\langle\cos^2\theta\rangle_0 = 1 - 2L/\xi$; tensor (10)	algebraic identity $L^2 + L' = 1 - 2L/\xi$ proved; trace = 1; both limits; numeric quadrature to 10^{-6}
FP equation (14)	annihilates Gibbs state; conserves norm; isotropy under pure $\boldsymbol{\Omega}$
Exact moment eq. (19)	all three operator identities verified by index algebra; equilibrium stationarity reproduced exactly
Debye eq. (20)	obtained as the exact $O(\xi)$ limit of (19)
PRL (6)	derived from (19) + Gibbs closure; coefficient-by-coefficient match ($\mathbf{h}$ and $\hat{\boldsymbol{\xi}}_e$ projections)
$\tau_\parallel, \tau_\perp$	both $\to \tau_B$ at $\xi \to 0$; strong-field behavior physical
$I/\tau_s = 6\eta\varphi$; factor $\frac{1}{2}$ in $\frac{1}{2}\nabla\times(\mathbf{M}\times\mathbf{H})$	direct evaluation; total angular momentum conservation
PRL (1),(3),(4)	inertia estimate $\tau_s/\tau_B \sim 10^{-5}$; substitution algebra
$\Delta\eta = \frac{3}{2}\eta\varphi g$; PRL (2)	two independent proofs (stress grouping; SM average) agree
Sh-72 static formula (SM 2)	full derivation; tanh identity checked; both limits
MRSh static formula (35)	full derivation; agrees with Sh-72 at both limits; tabulated difference at intermediate ξ
SM (23)/(32) M_y	derived two ways (circular decomposition vs. phasor); identical
SM (28) with conventions	quasi-static factor check ($\frac{1}{8}$ amplitude vs. $\frac{1}{4}$ rms); numeric ODE check at $\xi_0 = 0.3$ ($< 1\%$)
$\omega_m\tau_B = \sqrt{3}$, depth $-1/8$	R' computed; $R(\sqrt{3}) = -0.125$ numerically
SM (29) tail	prefactor re-derived from $\bar{\xi}/\tau_B$
SM (36),(37),(38); PRL (2) at all amplitudes	derived; weak-field limit reproduces Sec. 9
SM (40)	derived from scratch (Sec. 10.3); reproduces SM's quoted 0.98, 1.707, -0.105 exactly; scanned formula shown inconsistent (missing square)
PRL (7),(8),(9)	line-by-line derivations; static and weak-field limits recovered; regular (ξ_e, P) form given
PRL Fig. 3 curves	independent RK4 integration; table in Sec. 12 matches published minima/zeros
Conversions 5.25×10^{-3}, 10^{-2}	recomputed from m, kT, τ_B
-19 cP, 25%, $-10/3$ isoline, $\xi_c, \omega_c\tau_B$	arithmetic cross-checks against Figs. 2,4,5

16 Errata found during verification

1. **PRL Eq. (3):** printed $-\nabla\rho$; should be $-\nabla p$.
2. **SM Eq. (40) as scanned:** "$2 - \tanh\varepsilon - 2\tanh\varepsilon/\varepsilon$" is inconsistent with SM's own quoted numbers ($\omega_0\tau_B = 0.98$, $\omega_m\tau_B = 1.707$, -0.105); the correct expression, re-derived here, is $2 - \tanh^2\varepsilon - 2\tanh\varepsilon/\varepsilon$ (a squared tanh lost in print/scan).
3. **Previous version of these notes:** the weak-field AC result was written as $\Delta\eta = \frac{1}{4}\eta\varphi\xi_0^2 R$ with the *amplitude* ξ_0 while simultaneously $g = \xi_0^2 R/12$ — mutually inconsistent by a factor 2. Correct: $\Delta\eta = \frac{1}{8}\eta\varphi\xi_0^2 R = \frac{1}{4}\eta\varphi\bar{\xi}^2 R$, $g = \xi_0^2 R/12$. Confirmed by the quasi-static limit and by direct numerics (Sec. 12).

17 Open problems for follow-up work

- **Aggregation/polydispersity.** Redo $g(\xi_0, \omega\tau_B)$ averaged over a size distribution (e.g. the Gamma law used by Shliomis–Stepanov) with $\tau_B(V) \propto V$ and $m(V) \propto V$: does the hybrid

parametrization of the PRL emerge naturally from a broad distribution?

- **Beyond linear order in $\Omega\tau_B$:** shear-rate dependence of g (magnetically induced non-Newtonian behavior) at $\Omega\tau_B \sim 1$.
- **Field geometry:** general β, rotating/elliptic polarization within the MRSh closure (rotating fields transfer angular momentum at *all* frequencies).
- **Exact FP benchmark:** spectral solution of (14) (expansion in spherical harmonics) to quantify the residual error of the MRSh closure in exactly the PRL's $(\xi_0, \omega\tau_B)$ window, and to settle the Sh-72/MRSh discrepancy of Sec. 8.
- **Stability with MRSh:** the $\eta_{\text{total}} = 0$ neutral curve ($\xi_c = 126$, $\omega_c\tau_B = 20$) was computed with the Debye closure; recompute with (PRL 6).